Student Number: ____________________

2023
Higher School Certificate
Trial Examination

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black/blue pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For Section II, all relevant mathematical reasoning and calculations must be shown
- Write your student number and/or name at the top of every page

Total marks – 100

Section I – Pages 3 – 9
10 marks
- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – Pages 10 – 31
90 marks
- Attempt Questions 11 – 32
- Allow about 2 hour and 45 minutes for this section

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2023
Higher School Certificate
Trial Examination

Mathematics Advanced

Booklet 1 (21 marks)
Questions 1 – 13

Student Number: ______________________________

Section I (10 marks)

Attempts Questions 1 – 10

Allow about 15 minutes for this section.

Use the multiple-choice grid on page 9 for questions 1 – 10.

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1 If $\log_{\mathrm{b}} 3 = x$ and $\log_{\mathrm{b}} 4 = y$, which expression is equivalent to $\log_{\mathrm{b}}(36b)$?

(A) $2xy$

(B) $2xyb$

(C) $2x + y + 1$

(D) $x^2 + y + b$

2 The table below shows the values of the functions $f(x)$ and $g(x)$ for various values of x.

x	1	2	3	4	5
$f(x)$	3	4	5	1	2
$g(x)$	5	3	2	1	4

What is the value of $f(g(3))$?

(A) 2

(B) 3

(C) 4

(D) 5

3 Which of the following is equivalent to $\frac{d}{dx}\log_2(x-5)$?

(A) $\dfrac{1}{x \ln 2}$

(B) $\dfrac{1}{(x-5)\ln 2}$

(C) $\dfrac{2}{x}$

(D) $\dfrac{2}{x-5}$

4 A graph, $g(x)$, is obtained by transforming $f(x)$ such that $g(x) = -\frac{1}{2}f(x-1)+2$.

The point $(3, 4)$ is on the graph of $f(x)$. Where does this point correspond to on $g(x)$?

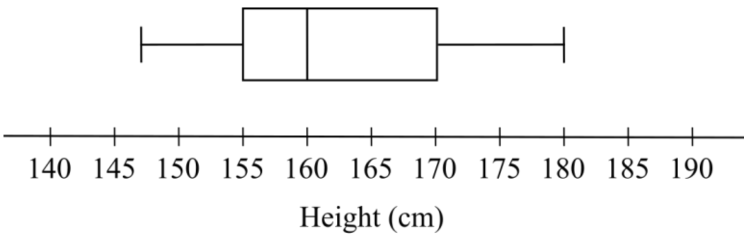
(A) $(4, 0)$

(B) $(2, 0)$

(C) $(-2, 6)$

(D) $(-1, 6)$

5 The heights of students in a class are represented in the boxplot below.



A new student, who will become the tallest member, is joining the class. What is the maximum height the new student can be in order to not be classified as an outlier?

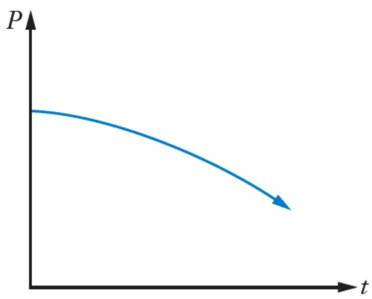
(A) 187.5 cm

(B) 192.5 cm

(C) 197.5 cm

(D) 202.5 cm

6 The population (P) of Marvel, a super villain's town is shown over time (t) by the graph below:



Which statement best describes the population of Marvel?

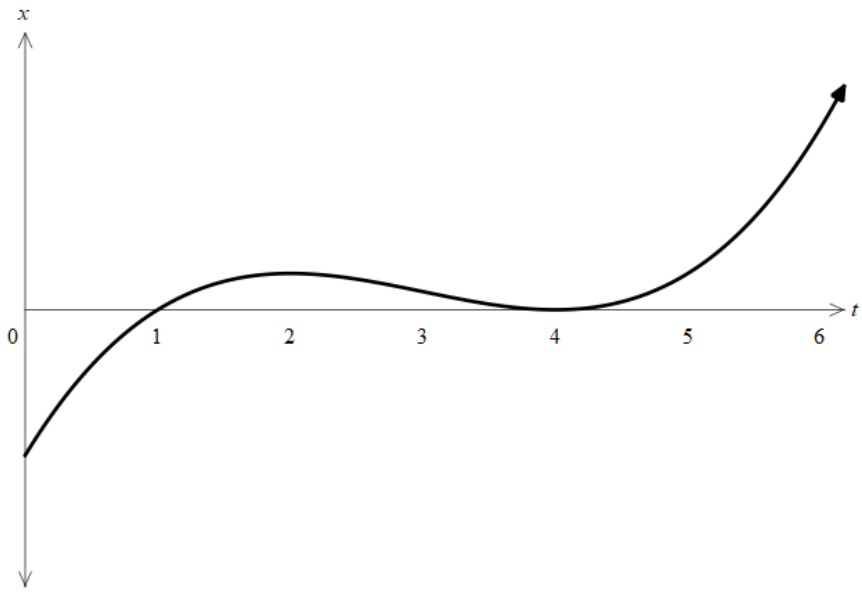
(A) The population is increasing at an increasing rate.

(B) The population is increasing at a decreasing rate.

(C) The population is decreasing at an increasing rate.

(D) The population is decreasing at a decreasing rate.

7 The displacement, x metres, from the origin of a particle moving in a straight line at any time, t seconds, is shown in the graph below.



When was the particle at rest?

(A) $t = 0$

(B) $t = 1$ and $t = 4$

(C) $t = 2$ and $t = 4$

(D) $t = 1, t = 2$ and $t = 4$

8 Find the range of $y = \frac{1}{2} - \frac{3}{2}\sin\left(2x + \frac{\pi}{3}\right)$

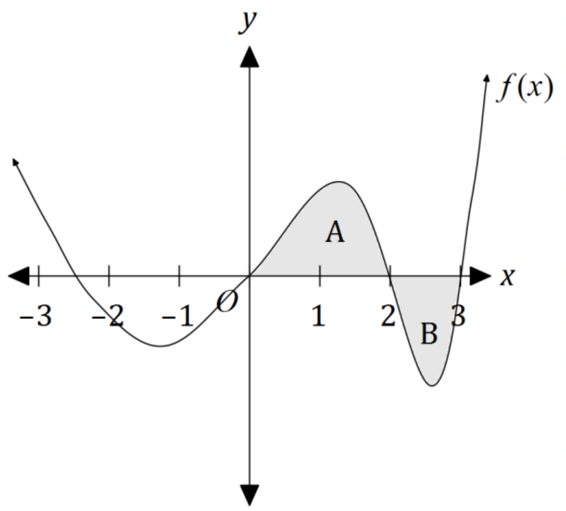
(A) $\left[-\frac{3}{2}, \frac{3}{2}\right]$

(B) $[-1, 2]$

(C) $[-1, 1]$

(D) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

9 The graph of a function $f(x)$ is shown below, where the shaded area labelled A is $\frac{3}{2}$ of the shaded area labelled B.



If $\int_0^3 f(x)dx = 2$, what is the area of A?

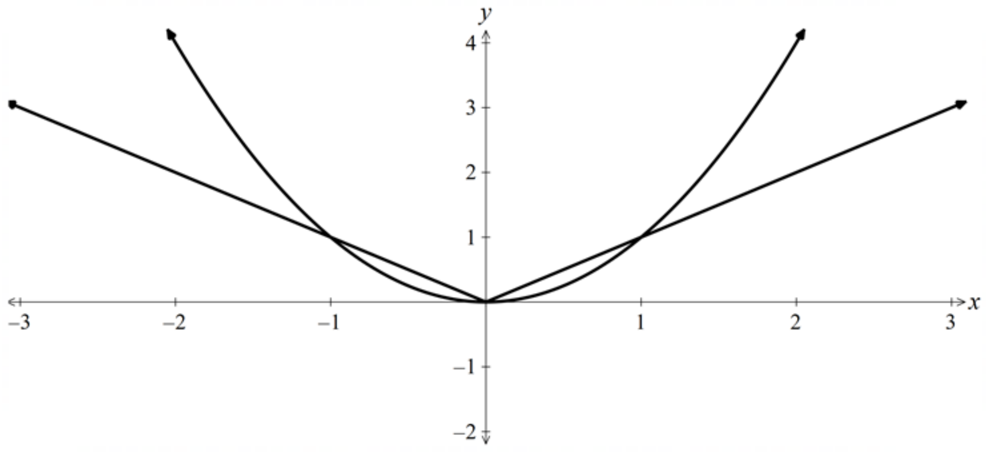
(A) 6

(B) 4

(C) $\frac{6}{5}$

(D) $\frac{4}{5}$

10 The graph shows $y = x^2$ and $y = |x|$.



What is the solution to the inequality $x^2 - |x| > 0$?

(A) $-1 \leq x \leq 1$

(B) $-1 < x < 1$

(C) $x \leq -1 \ \ and \ \ x \geq 1$

(D) $x < -1 \ \ and \ \ x > 1$

End of Section I

Record your answers below by shading the appropriate box.

1	**A**	**B**	**C**	**D**
2	**A**	**B**	**C**	**D**
3	**A**	**B**	**C**	**D**
4	**A**	**B**	**C**	**D**
5	**A**	**B**	**C**	**D**
6	**A**	**B**	**C**	**D**
7	**A**	**B**	**C**	**D**
8	**A**	**B**	**C**	**D**
9	**A**	**B**	**C**	**D**
10	**A**	**B**	**C**	**D**

2023
Higher School Certificate
Trial Examination

Mathematics Advanced
Section II

90 marks

Attempt Questions 11 – 32

Allow about 2 hours and 45 minutes for this section

Instructions

- Answer each question in the space provided.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over

MARKS

Question 11 (5 marks)

Buns & Roses sells bread rolls for \$0.80 each. The cost, C dollars, of making n bread rolls is given by $C = 300 + 0.2n$.

(a) How much does every 100 bread rolls made add to the cost of production? **1**

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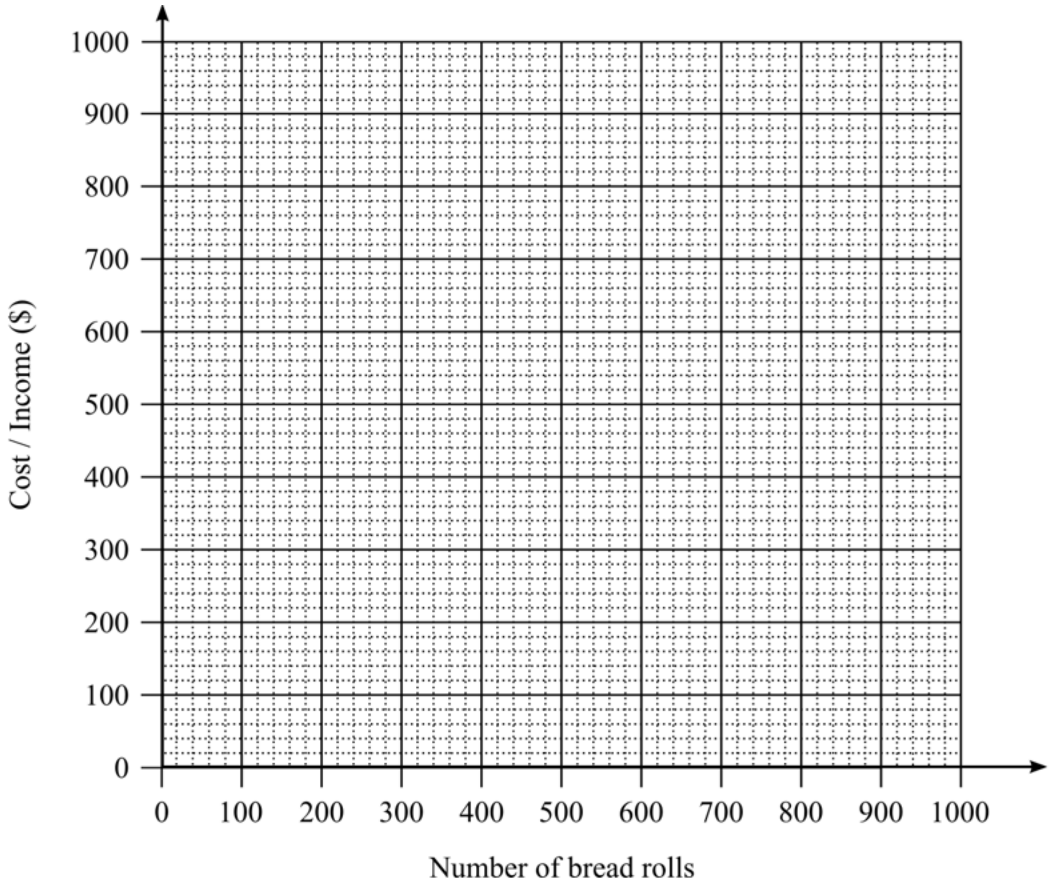
(b) On the grid below, draw the graphs of the cost, C, and the income, I **and** hence, find the number of bread rolls that must be sold to break even. **2**



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(c) How many bread rolls must be sold to make a profit of \$240? **2**

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Question 12 (4 marks)

Let $f(x) = \frac{x}{x^2-1}$.

(a) Find $f'(x)$. **2**

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(b) Hence or otherwise, find the equation of the normal to the graph of $y = f(x)$ at the point where $x = 2$. **2**

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Question 13 (2 marks)

Differentiate $5x^2 \tan\left(\frac{1}{x}\right)$ **2**

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End of Booklet 1

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

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2023
Higher School Certificate
Trial Examination

Mathematics Advanced

Booklet 2 (39 marks)
Questions 14 – 25

Student Number: ____________________________

MARKS

Question 14 (2 marks)

Use two applications of the trapezoidal rule to find an approximation of the value of the integral $\int_0^2 e^{-\frac{x^2}{2}} dx$. Give your answer correct to 2 decimal places. **2**

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Question 15 (2 marks)

Solve $|2x-3|=9$ **2**

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Question 16 (4 marks)

(a) Find $\int \frac{x+1}{x^2+2x+1} dx$ **2**

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(b) Evaluate $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sec^2 \frac{x}{2} dx$ **2**

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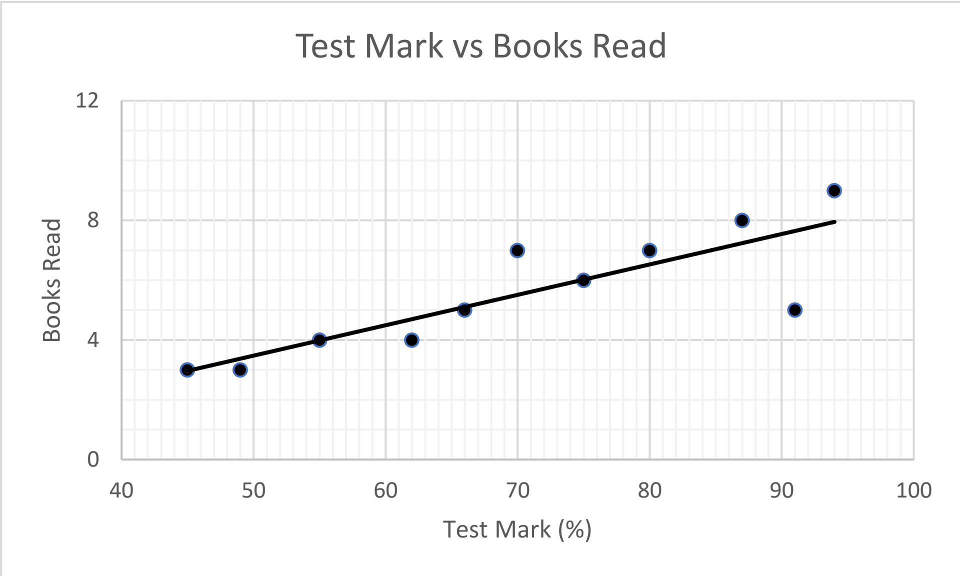
Question 17 (3 marks)

A group of Year 6 students were surveyed about their reading habits.

The students were then given a reading comprehension test.

The scatterplot below shows the results of the test plotted against the number of books that the student had read in the previous month.



The line of best fit for the data has also been added.

(a) How many students were part of the survey? **1**

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(b) Pearson's correlation coefficient for this scatterplot is 0.843. **1**

What can be concluded about the correlation between the two variables?

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(c) A student who was not part of the survey has read 7 books in the past month. **1**

Predict the mark that this student would get if they sat the reading comprehension test.

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Question 18 (3 marks)

Let there be two events, A and B, such that $P(A) = 0.4$, $P(B) = 0.55$ and $P(A \cup B) = 0.8$.

(a) Find the value of $P(\overline{A} \cap \overline{B})$. **1**

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(b) Hence, or otherwise, find the value of $P(A|\overline{B})$. **2**

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Question 19 (3 marks)

Solve $2sin^2x - 5sinxcosx + 3cos^2x = 0$, for $x \in [0,2\pi]$. **3**

Leave your answer in exact form.

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Question 20 (6 marks)

Since the introduction of the internet, the number of emails sent has been modelled using the exponential equation

$$M = M_0 e^{kt}$$

where M is the number of emails sent worldwide, in millions, and t is the number of years after the year 1980. Records show that 45000 emails were sent in 1980 and 306 billion emails were sent in the year 2020.

(a) Find the values of M_0 and k, correct to three decimal places. **2**

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(b) How many emails, to the nearest million, are expected to be sent in the year 2025? **2**

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(c) Find the rate at which the number of emails sent increased in the year 2000, correct to the nearest million per year. **2**

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Question 21 (2 marks)

The table below shows the probability distribution of a discrete random variable X.

x	0	2	4	5	8	9
$P(X=x)$	k^2	0.16	0.18	0.3	k	0.12

(a) Show that $k=0.2$. **1**

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(b) Calculate $E(X)$. **1**

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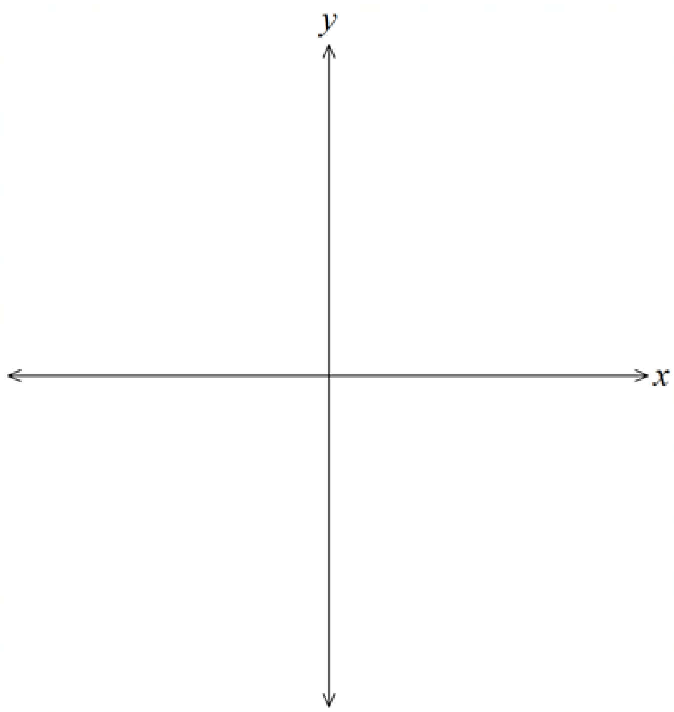
Question 22 (3 marks)

For a given function, $f(x)$, it is known that $f(a)=f(-a)$.

Also $f'(2)=f'(0)=0$ and $f'(x)<0$ for $0<x<2$ and $f'(x)>0$ for $x>2$.

Sketch a possible curve for $y=f(x)$ if it passes through the origin.



Question 23 (3 marks)

The table below shows future value factors for an annuity of $1. **3**

	A	B	C	D	E	F	G
1	**Table of future value interest factors**						
2	Periods\Rate	Interest rate per period					
3		0.50%	1.00%	1.50%	2.00%	3.00%	6.00%
4	5	5.0503	5.1010	5.1523	5.2040	5.3091	5.6371
5	10	10.2280	10.4622	10.7027	10.9497	11.4639	13.1808
6	20	20.9791	22.0190	23.1237	24.2974	26.8704	36.7856
7	40	44.1588	48.8864	54.2679	60.4020	75.4013	154.7620
8	60	69.7700	81.6697	96.2147	114.0515	163.0534	533.1282

Wayne and Vicki each undertake an annuity at 6% p.a. for a period of 5 years.

- Wayne invests $3000 per quarter, compounded quarterly.
- Vicki invests $1000 per month, compounded monthly.

Using supporting calculations, determine who will earn the greater amount of interest over the 5 years.

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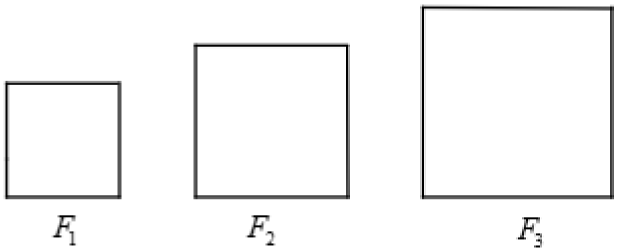
Question 24 (3 marks)

In the diagram below, F_1, F_2, F_3 are square frames. The perimeter of the first frame is 16cm. The perimeter of each square frame afterwards is 4cm longer than the perimeter of the previous frame.



(a) Find the perimeter of F_{12}. **1**

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(b) A thin metal wire of length 2000cm is cut into pieces and these pieces are then bent to form the above square frames. **2**

Find the greatest number of distinct square frames that can be formed.

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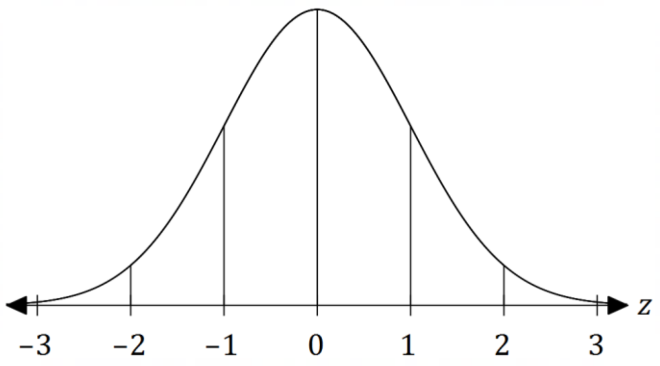
Question 25 (5 marks)

The weights of newborn babies are normally distributed with a mean of 7.5 pounds and a standard deviation of 1.1 pounds.

(a) A normal distribution curve is shown below, where the vertical lines represent z-scores from -3 to 3. **1**



A baby is considered underweight if their birthweight has a z-score of less than -1.8. **Shade** the area under the normal distribution curve representing the percentage of babies that are born underweight.

(b) Below what birthweight, in pounds, is a baby considered to be underweight? **2**

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(c) If the area under the normal distribution curve below $z = 1.8$ is equal to 0.9641, how many babies in 1,000 would be expected to be born underweight? **2**

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End of Booklet 2

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

2023
Higher School Certificate
Trial Examination

Mathematics Advanced

Booklet 3 (40 marks)
Questions 26 – 32

Student Number: ____________________________

MARKS

Question 26 (7 marks)

As a particle moves, its velocity, in metres per second, is described by the equation

$$v(t) = -2t^2 + 2t + 4$$

where $t \geq 0$ is the time in seconds. The particle is initially 4 metres right of the origin.

(a) Find the time taken for the particle to reach its maximum velocity. **2**

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(b) Find the position of the particle when it first comes to rest. **3**

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(c) Find the total distance travelled by the particle in the first 3 seconds. **2**

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Question 27 (6 marks)

Julio has just retired and invests part of his superannuation into an annuity.
He invests \$400 000 into an account that earns interest at 3% p.a. and receives regular monthly payments of \$$M$ from the annuity at the end of each month.

(a) Show that the amount remaining in the annuity after the third monthly payment is given by: **2**

$$A_3 = 400\,000 \times 1.0025^3 - M(1 + 1.0025 + 1.0025^2)$$

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(b) Determine the monthly payment that Julio receives given that the annuity will last for 25 years. **4**

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Question 28 (7 marks)

Let $f(x) = x^3 + x^2 - x$.

(a) Find the stationary points on the graph of $y = f(x)$ and determine their nature. **3**

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(b) Hence, sketch the graph of $y = f(2x)$. *Only* label the turning points and the y-intercept. **2**

(c) Find the possible values of a such that the graph of $y = f(2x) + a$ has only one x-intercept. **2**

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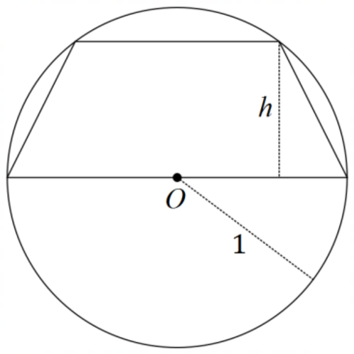
Question 29 (7 marks)

A trapezium is inscribed in a circle centred at O with a radius of 1, such that one of its bases is a diameter of the circle, as shown in the diagram below. The height of the trapezium, h, may vary.



(a) Show that the area of the trapezium can be given by $A = h\sqrt{1-h^2} + h$. **2**

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(b) Find the exact value of the maximum area of the trapezium. **5**

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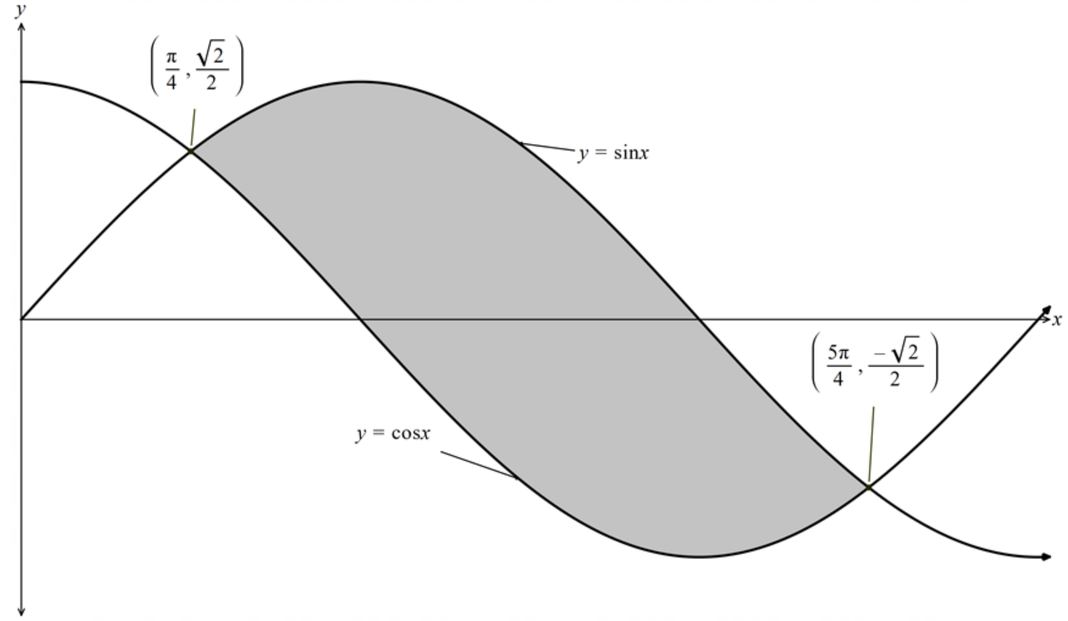
Question 30 (3 marks)

The graph below shows an area enclosed between the curves $y = \sin x$ and $y = \cos x$ **3**



Find an exact value for the shaded area.

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Question 31 (6 marks)

A continuous random variable, X, is defined by the probability density function:

$$P(X) = \frac{3\sqrt{x-3}}{16}, \qquad 3 \leq x \leq k$$

(a) Show that $k = 7$. **2**

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(b) Determine the median value of X, correct to two decimal places. **2**

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(c) Using appropriate mathematical concepts, explain why the mode of X is 7. **2**

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Question 32 (4 marks)

Two surveyors, A and B, stand on level ground to observe a large tree. The tree is directly north of surveyor A who sights the top of the tree at an angle of elevation of $52°$. Surveyor B is 27 metres away from Surveyor A on a bearing of $043°$ and sights the top of the tree at an angle of elevation of $46°$. **4**

Find the height, h, of the tree correct to the nearest metre.

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End of Booklet 3

End of Examination

This space can be used if needed. Ensure that you label any answers written here clearly giving their question number and part letter.

NSW GOVERNMENT

NSW Education Standards Authority

2021 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

and $\alpha\beta\gamma = -\frac{d}{a}$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, \ r \neq 1$$

$$S = \frac{a}{1 - r}, \ |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

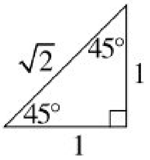
$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

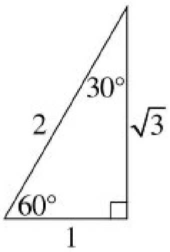


$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \ \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \ \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \ \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

If $t = \tan\frac{A}{2}$ then $\sin A = \frac{2t}{1+t^2}$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A\cos B = \frac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$$

$$\sin A\sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$$

$$\sin A\cos B = \frac{1}{2}\left[\sin(A + B) + \sin(A - B)\right]$$

$$\cos A\sin B = \frac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

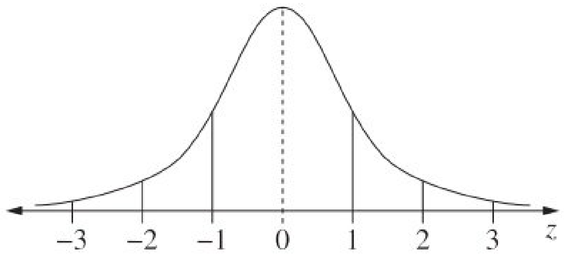
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between –1 and 1
- approximately 95% of scores have z-scores between –2 and 2
- approximately 99.7% of scores have z-scores between –3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E\left[(X - \mu)^2\right] = E\left(X^2\right) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \ P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x)\,dx$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow \ P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \ x = 0, 1, \ldots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\dfrac{dy}{dx} = nf'(x)[f(x)]^{n-1}$
$y = uv$	$\dfrac{dy}{dx} = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$
$y = \dfrac{u}{v}$	$\dfrac{dy}{dx} = \dfrac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\dfrac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\dfrac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\dfrac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\dfrac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\dfrac{dy}{dx} = (\ln a)f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{(\ln a)f(x)}$
$y = \sin^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \cos^{-1} f(x)$	$\dfrac{dy}{dx} = -\dfrac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \tan^{-1} f(x)$	$\dfrac{dy}{dx} = \dfrac{f'(x)}{1+[f(x)]^2}$

Integral Calculus

$$\int f'(x)[f(x)]^n\,dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x)\sin f(x)\,dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}\,dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + c$$

$$\int f'(x)a^{f(x)}\,dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}\,dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2+[f(x)]^2}\,dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$$

$$\int_a^b f(x)\,dx$$

$$\approx \frac{b-a}{2n}\left\{f(a) + f(b) + 2\left[f(x_1) + \cdots + f(x_{n-1})\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$\left|\underset{\sim}{u}\right| = \left|x\underset{\sim}{i} + y\underset{\sim}{j}\right| = \sqrt{x^2 + y^2}$$

$$\underset{\sim}{u} \cdot \underset{\sim}{v} = \left|\underset{\sim}{u}\right|\left|\underset{\sim}{v}\right|\cos\theta = x_1x_2 + y_1y_2,$$

where $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$

and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos\theta + i\sin\theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} \left[r(\cos\theta + i\sin\theta)\right]^n &= r^n(\cos n\theta + i\sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$x = a\cos(nt + \alpha) + c$$

$$x = a\sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$